

Nanjing University of Information Science and Technology

Laboratory Report

Experiment Title: Cloud-Native Automated Deployment

Date	2026-06-10	Class	2
Name	whatokk	Student ID	
GitHub Account	whatokk	Teacher	yhldrf / yhldrf@gmail.com
Repository	whatokk/cloud-computing-lab-github	Live URL	https://whatokk.github.io/cloud-computing-lab-github/

一、Experimental Purpose

The purpose of this experiment is to practice a cloud-native automated deployment workflow using GitHub as a PaaS-like platform. The experiment covers forking a cloud repository, enabling GitHub Pages, triggering GitHub Actions, verifying a globally accessible website, and inviting the teacher as a collaborator.

二、Experiment Contents

- Forked/acquired the cloud computing lab repository.
- Enabled GitHub Pages with GitHub Actions as the build and deployment source.
- Personalized index.html by adding the student name, UI style changes, and an external image.
- Triggered the Cloud-Deployment-Pipeline and verified that the workflow completed successfully.
- Opened and verified the live GitHub Pages URL.
- Invited teacher yhldrf as a collaborator with write permission.

三、Detailed Steps and Results

Step 1 Resource Acquisition

Logged in to GitHub and used the lab source repository <https://github.com/yhldrf/cloud-computing-lab-github>. Because a previous same-name fork already existed in the account, the existing fork was updated for Lab6 while preserving the separate report-submission branch.

Step 2 Enabling the Hosting Service

Configured GitHub Pages to use workflow deployment. The Pages API returned `build_type=workflow` and the live site URL.

Step 3 Triggering Cloud Automation

Edited `index.html`: replaced the placeholder name with `whatokk`, changed the background color and status color, and added a GitHub Actions image. The changes were committed and pushed to `main`, then the Cloud-Deployment-Pipeline workflow was triggered.

Step 4 Verification and Global Access

The workflow completed with `conclusion=success`. The website was available at <https://whatokk.github.io/cloud-computing-lab-github/>, and the downloaded live HTML contains Student Name: `whatokk` and Site Status: `Live`.

Step 5 Content Personalization

The UI style was updated to a light gray background, the status text was changed to blue, and an external SVG image was included using an `img` tag.

Collaborator

Teacher account `yhldrf` was invited as collaborator with write permission. GitHub creates an invitation that the teacher can accept from GitHub.

四、Evidence Screenshots

Resource Acquisition and Pages Service

Captured on Windows 10 | 2026-06-10 05:46:08 +08:00 | GitHub account: whatokk

```
Source repository: https://github.com/yhldrf/cloud-computing-lab-github
Working fork/repository: https://github.com/whatokk/cloud-computing-lab-github
Pages build type: workflow
Live URL: https://whatokk.github.io/cloud-computing-lab-github/
HTTPS enforced: True
```

Figure 1. Repository and GitHub Pages configuration.

```
Cloud Automation Evidence

Captured on Windows 10 | 2026-06-10 05:46:08 +08:00 | GitHub account: whatokk

Workflow: Cloud-Deployment-Pipeline
Event: workflow_dispatch
Status: completed
Conclusion: success
Run URL: https://github.com/whatokk/cloud-computing-lab-github/actions/runs/27237668861
deploy Checkout Code 2026-06-09T21:41:31.8376747Z ##[group]Run actions/checkout@v4
deploy Setup Cloud Environment 2026-06-09T21:41:32.3425691Z ##[group]Run actions/configure-pages@v4
deploy Upload Artifact to Cloud Storage 2026-06-09T21:41:32.5804220Z ##[group]Run actions/upload-pages-artifact@v3
deploy Execute Deployment 2026-06-09T21:41:33.2383078Z ##[group]Run actions/deploy-pages@v4
deploy Set up job 2026-06-09T21:41:31.7524388Z Complete job name: deploy
```

Figure 2. GitHub Actions deployment pipeline result.

```
Live Site Verification Evidence

Captured on Windows 10 | 2026-06-10 05:46:08 +08:00 | GitHub account: whatokk

body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; text-align: center; padding-top: 50px;
background-color: #f4f4f9; }
.status { color: #007bff; font-weight: bold; }
<h1>Cloud-Native Deployment Success!</h1>
<p>This site was deployed using <b>GitHub Actions</b>.</p>
<div class="status">● Site Status: Live</div>
<p>Student Name: whatokk</p>
<p></p>
```

Figure 3. Live site verification and personalized HTML content.

```
Collaborator Invitation Evidence

Captured on Windows 10 | 2026-06-10 05:46:08 +08:00 | GitHub account: whatokk

Repository: https://github.com/whatokk/cloud-computing-lab-github
Teacher GitHub user: yhldrf
Teacher email provided: yhldrf@gmail.com
Invitation permissions: write
Invitation URL: https://github.com/whatokk/cloud-computing-lab-github/invitations
Created at: 2026-06-09T21:44:24Z
```

Figure 4. Collaborator invitation evidence.

五、Reflection

Where is the website stored? The source files are stored in the GitHub repository, and the deployed static website is served through GitHub Pages using artifacts produced by GitHub Actions.

Did you have to install any software? The cloud deployment itself did not require installing server software. GitHub provided the repository hosting, Ubuntu runner, deployment workflow, and Pages hosting service. Local GitHub CLI was only used to automate the same operations more efficiently.

Problems Encountered and Proposed Solutions: The account already had a same-name fork from another lab submission, so a clean Lab6 main branch was reconstructed from the teacher source repository and the previous PR branch was left untouched. The workflow was then manually enabled and dispatched to guarantee a verifiable run record.

六、Links

Teacher source repository: <https://github.com/yhldrf/cloud-computing-lab-github>

Working repository: <https://github.com/whatokk/cloud-computing-lab-github>

Successful Actions run: <https://github.com/whatokk/cloud-computing-lab-github/actions/runs/27237668861>

Live GitHub Pages site: <https://whatokk.github.io/cloud-computing-lab-github/>